

WORK EXPERIENCE

Qualcomm (formerly Alphawave SEMI)

Toronto, Canada

Physical Design Engineer (PEY)

May 2025 - Present

- Automated and standardized a **Fusion Compiler** physical design flow, introducing guardrails, regression and trial tooling, and user-safe interfaces to improve run reliability and scalability across teams.
- Integrated **PrimeTime** for **DMSA timing analysis** and **ECO fixing**, enabling robust multi-corner closure and improved post-route timing convergence.
- Built scalable QoR infrastructure and **LLM-powered RAG agents** to collect, analyze, and query timing, power, utilization, and DRC metrics, enabling natural-language QoR review, trend analysis, and actionable next-step recommendations.
- Designed and deployed a cross-team design handoff system to manage flow collateral, ensuring version consistency, traceability, and reproducibility across the ASIC development lifecycle.

Bank of Montreal (BMO)

Toronto, Canada

Junior Software Developer

May 2024 - August 2024

- Developed and implemented an **online self-serve platform** allowing users to deploy and manage **Kafka topics, access controls, and data schemas**, removing the need for a CLI, Angular, and Node.js
- Created and optimized **Python scripts** to proactively monitor **Apache Kafka connectors**, automating **real-time alerts** and enhancing operational efficiency across the team
- Played a pivotal role in **migrating Kafka streams from EC2 to ECS**, leading to a substantial boost in **system performance** and significant **cost savings**

Expert Geophysics Limited (EGL)

Aurora, Canada

Engineering Intern

May 2023 - August 2023

- Developed and tested **C-based software** for **airborne magnetic field monitoring systems**, ensuring optimal performance through rigorous design and testing in **critical operations**
- Conducted testing on key components including the **console, current generator, and power supply**, ensuring **system reliability and functionality**
- Analyzed and improved the **structural integrity of magnetometer mounts**, implementing **3D-printed designs** that increased the **safety factor by 33%** and reduced **maximum displacement by 83%**

PROJECTS

BMO Innov8 Hackathon — Angular, Node.js, Apache Kafka, Figma

- Won 1st place** at the **prestigious BMO Innov8 Hackathon**, a competitive event with **over 300 participants** and **30 teams**
- Developed an innovative solution for a financial technology challenge, showcasing advanced problem-solving and technical skills

EDUCATION

University of Toronto — St. George

B.A.Sc in Computer Engineering + PEY

September 2022 - June 2026 GPA: 3.64

- Data Structures and Algorithms, Operating Systems, Computer Networks, Programming Fundamentals (C++), Software Communication and Design (C++)

SKILLS AND CERTIFICATIONS

- Certifications** - Amazon Web Services Certified Cloud Practitioner — October 2021 – September 2027
- Programming Languages** - Python, C++, C, Java, JS/TS, SQL, TCL, Bash, Perl, MATLAB, Assembly
- EDA & HW Tools** - Fusion Compiler, PrimeTime, StarRC, IC Validator (ICV), Formality (FM), Verilog
- Technologies** - Linux, Git, SVN, SLURM, Streamlit, LangChain, Ollama, Docker, Apache Kafka, AWS, Google Cloud Platform, NumPy, Pandas, Matplotlib, YAML, JSON, Makefiles
- Competencies** - Physical Design (PnR), Static Timing Analysis (STA), Timing Closure, QoR Analysis, Scripting & Automation